DeFi & Advanced Crypto

Module 3 Study Guide

Yield Farming & Staking

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: Staking Fundamentals

In this lesson: Network Consensus; Three Staking Models; Slashing & Lock-ups; Worked TT$ Ex

Key points

- Direct staking offers full control with technical operational risk
- Delegated staking simplifies hardware but requires careful validator vetting
- Liquid staking unlocks frozen funds while unbonding periods govern exits

Examples from the lesson

- This is Module 3, Yield Farming and Staking, and we are kicking things off with Lesson 1: Staking Fundamentals.
- There are three primary ways you can participate in staking, starting with direct Proof-of-Stake staking.
- Liquid staking protocols give you a derivative token in exchange for your staked assets, representing your deposit 1:1.
- Imagine you have TT$ 1,000 worth of cryptocurrency that you decide to deposit into a liquid staking protocol.
- The protocol immediately issues you liquid derivative tokens worth TT$ 1,000, allowing you to maintain full financial flexibility across other DeFi applications while your principal remains actively staked.
- Now suppose the external validator you selected experiences an unexpected server failure and suffers a minor five percent network penalty due to slashing.
- A five percent penalty on your original TT$ 1,000 stake equals a reduction of TT$ 50.
- This brings your underlying staked balance down to TT$ 950, demonstrating why choosing high-uptime, reliable validators is far more important than chasing unverified network claims.
- If you decide to convert your liquid tokens back into the original asset, the protocol enforces a strict seven-day lock-up period where your funds remain illiquid.
- During these seven days, your funds cannot be moved or traded, protecting network stability while the system processes your unstacking request.
- In our next lesson, Lesson 2: Liquidity Pools and Impermanent Loss, we will explore how to supply pairs of assets to decentralized exchanges and calculate the unique risks involved.

Lesson 2: Liquidity Pools and Impermanent Loss

In this lesson: Define liquidity pools; Explain constant product; Work TT$ impermanent loss ex

Key points

- Defined liquidity pools and constant product rule
- Walked through TT$10,000+TT$10,000 example showing ~2% loss
- Discussed when fees outweigh impermanent loss

Examples from the lesson

- You are in the DeFi & Advanced Crypto course, Module 3, Lesson 2: Liquidity Pools and Impermanent Loss.
- A liquidity pool is a smart contract that holds two tokens, enabling traders to swap between them without a traditional order book.
- Suppose you deposit TT$10,000 worth of Token X and TT$10,000 worth of Token Y into a 50/50 pool, so the initial price is 1 X = 1 Y.
- The pool now contains 10,000 X and 10,000 Y.
- After some trading, the market price of X rises to 1.5 Y while Y stays at 1 Y.
- Using the constant product, the new balances become approximately 8,165 X and 12,247 Y.
- Your pool share is worth (8,165 × 1.5) + 12,247 = 24,495 TT$.
- If you had simply held the original tokens, their value would be (10,000 × 1.5) + 10,000 = 25,000 TT$.
- The impermanent loss is (24,495 − 25,000) / 25,000 ≈ -2.02 %, or about TT$405 loss on your TT$20,000 position.
- Notice the loss is “impermanent” because if the price returns to 1:1, your pool share would regain its original value.
- We defined liquidity pools and the constant product rule, explained impermanent loss, walked through a TT$10,000 + TT$10,000 example showing roughly a 2 % loss when one token’s price rose 50 %, and discussed how fees and holding period affect whether liquidity provision is sensible.
- In the next lesson, Lesson 3: Evaluating Yield Without Chasing It, we’ll learn how to assess real returns and avoid the temptation of chasing high-yield scams.

Lesson 3: Evaluating Yield Without Chasing It

In this lesson: Origin of Yield; Real vs Incentive Yield; Yield Audit Checklist; Protecting Your Ca

Key points

- Real yield stems from trading fees, not token printing
- Token incentives introduce heavy price collapse risks
- Always audit cash flow origin and token liquidity

Examples from the lesson

- We are currently in Module 3, "Yield Farming and Staking," and today we are covering Lesson 3: "Evaluating Yield Without Chasing It."
- Imagine you put TT$10,000 into a DeFi liquidity pool or staking protocol to test out an advertised yield offer.
- Over the course of one full year, the pool generates TT$400 in actual trading fees paid by users swapping assets.
- That TT$400 represents a sustainable, real yield of 4% based on genuine platform usage.
- However, to make the offer look irresistible, the protocol also prints and hands you TT$1,600 worth of their native reward token.
- On your dashboard, that total payout looks like a massive TT$2,000 return, which gives you a headline yield of 20%.
- But here is the catch: that extra TT$1,600 in reward tokens relies entirely on new buyers entering the market to support the token's price.
- If market demand drops and holders sell off those reward tokens, that TT$1,600 portion can quickly collapse toward zero, leaving you with only the TT$400 in real fee revenue.
- A 20% headline yield made of 4% real fees and 16% token rewards carries high price risk.

Recap checklist — can you say each one?

Lesson 1: Staking Fundamentals

- ☐ Direct staking offers full control with technical operational risk
- ☐ Delegated staking simplifies hardware but requires careful validator vetting
- ☐ Liquid staking unlocks frozen funds while unbonding periods govern exits

Lesson 2: Liquidity Pools and Impermanent Loss

- ☐ Defined liquidity pools and constant product rule
- ☐ Walked through TT$10,000+TT$10,000 example showing ~2% loss
- ☐ Discussed when fees outweigh impermanent loss

Lesson 3: Evaluating Yield Without Chasing It

- ☐ Real yield stems from trading fees, not token printing
- ☐ Token incentives introduce heavy price collapse risks
- ☐ Always audit cash flow origin and token liquidity